Solution:

Problem 1.

Let

(1) $$A = \begin{bmatrix} 2 & -1 & 1 \\ 6 & 1 & 2 \\ 2 & -1 & 3 \end{bmatrix} \quad \&$$

(2) $$b = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}.$$

(a) Finding L & U s.t $A = LU$:

let me start row ops on $[A \mid b]$. We want to do row ops on the augmented matrix so that to check our result at the end:

$$e_{ip} = -\frac{r_{ip}}{r_{pp}}$$

← row i operation entry

→ pivot row entry

$$[A \mid b] = \left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 6 & 1 & 2 & 2 \\ 2 & -1 & 3 & 0 \end{array}\right] \xleftarrow[e_{21} = -\frac{6}{2} = -3]{R2 = r2 - 3r1} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 0 & 4 & -1 & -4 \\ 2 & -1 & 3 & 0 \end{array}\right]; \quad E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(3) $$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 0 & 4 & -1 & -4 \\ 2 & -1 & 3 & 0 \end{array}\right] \xleftarrow[e_{31} = -\frac{2}{2} = -1]{R3 = r3 - r1} \left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 0 & 4 & -1 & -4 \\ 0 & 0 & 2 & -2 \end{array}\right]; \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\underbrace{}_{\text{already } U} \qquad \underbrace{}_{c}$$

So we've come to the upper triangular matrix U with

$$E_{31} \cdot E_{21} [A \mid b] = [U \mid c] \quad \text{where}$$

(4) $$U = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & 0 & 2 \end{bmatrix} \quad \& \quad c = \begin{bmatrix} 2 \\ -4 \\ -2 \end{bmatrix}.$$

To obtain L we need:

(5) $$E_{31} \cdot E_{21} \cdot A = U \quad \text{where}$$

$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ & $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Since these E_{ij} are elementary matrices, they <u>do have</u> inverses E_{ij}^{-1} that undo what E_{ij}'s do. So they're invertible. Then

$$E_{31} \cdot E_{21} \cdot A = U \quad \Big\} * E_{31}^{-1} \text{ by left} \Rightarrow$$

$$\underbrace{(E_{31}^{-1} \cdot E_{31})}_{I} \cdot E_{21} \cdot A = E_{31}^{-1} \cdot U \quad \Rightarrow$$

$$E_{21} \cdot A = E_{31}^{-1} \cdot U \quad \Big\} * E_{21}^{-1} \text{ by left} \Rightarrow$$

(6) $$\boxed{A = E_{21}^{-1} \cdot E_{31}^{-1} \cdot U}$$

To calculat (6):

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}^{-1} \cdot U = \overbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}^{\text{Note: } E_{31}^{-1}} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \cdot U \Rightarrow$$

$$\Rightarrow A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}}_{\text{let } L} \cdot U := L \cdot U, \quad \text{so}$$

(7) $$\boxed{A = L \cdot U}, \quad \text{where } L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \text{ \& } U = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

(8) Use L & U found in (7) for solving (1):

$$\underbrace{Ax}_{\downarrow} = b \quad \text{can now be written}$$

$$\overbrace{LU}\,x = b \quad \Big\{ \cdot \overbrace{L^{-1}}^{\text{exist because } L \text{ is a product of invertible matrices}} \text{ of BS by left:}$$

$$L^{-1} \cdot L\,U\,x = L^{-1} \cdot b$$

(5) $$U\,x = L^{-1} b \quad \text{where by (2) \& (7)}$$

$$\underbrace{\begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & 0 & 2 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1}}_{L^{-1}} \cdot \underbrace{\begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}}_{b} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}$$

(6) $$\begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ -2 \end{bmatrix} \Rightarrow$$

← matching the c in (4) ✓

Solving this (6) by back substitution:

$$2x_3 = -2 \quad \Rightarrow \quad x_3 = -1$$
$$4x_2 - x_3 = -4 \quad \Rightarrow \quad x_2 = -5/4$$
$$2x_1 - x_2 + x_3 = 2 \quad \Rightarrow \quad x_1 = 7/8$$

(7) $$[x_1 \;\; x_2 \;\; x_3]^T = \left[\frac{7}{8}, \; \frac{-5}{4}, \; -1\right]^T$$

checking:

$$\begin{bmatrix} 2 & -1 & 1 \\ 6 & 1 & 2 \\ 2 & -1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 7/8 \\ -5/4 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{7}{4} + \frac{5}{4} - 1 \\ \frac{21}{4} - \frac{5}{4} - 2 \\ \frac{7}{4} + \frac{5}{4} - 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} \equiv b \quad \text{in (2). ✓.}$$